Exam Preparation: Data Mining & Warehousing

Time: 3 Hours (Estimated) **Max Marks:** 80

Section A: Short Answer Questions (3 Marks Each)

Instructions: Answer clearly and concisely. Bullet points are recommended for higher scoring.

1. Define Data Mining.

- **Definition:** Data mining is the process of discovering patterns, correlations, and anomalies within large datasets to predict outcomes.
- **Key Concept:** It transforms raw data into useful information using techniques from statistics, machine learning, and database systems.
- **Goal:** To extract knowledge (KDD) for decision-making.

2. Describe any two applications of Data Mining.

1. **Market Basket Analysis:** Retailers use this to identify which items are frequently bought together (e.g., bread and butter) to optimize store layout.
2. **Fraud Detection:** Financial institutions analyze transaction patterns to identify unusual activity that may indicate credit card fraud.

3. Discuss data preprocessing. Mention any one objective.

- **Discussion:** Data preprocessing is the technique of converting raw, messy data into a clean dataset. It involves cleaning, integration, transformation, and reduction.
- **Objective:** To improve data quality (accuracy and completeness) so that mining algorithms can work efficiently and produce reliable results.

4. Discuss about KDD (Knowledge Discovery in Databases).

- KDD is the overall process of converting raw data into useful knowledge.
- **Steps:** Selection $\rightarrow$ Preprocessing $\rightarrow$ Transformation $\rightarrow$ Data Mining $\rightarrow$ Interpretation/Evaluation.
- *Note:* Data mining is just one specific step within the broader KDD process.

5. Differentiate between Data Cleaning and Data Integration.

- **Data Cleaning:** Focuses on removing noise, filling missing values, and correcting inconsistencies (e.g., fixing "F" and "Female" to a standard format).
- **Data Integration:** Focuses on combining data from multiple sources (e.g., databases, flat files) into a coherent data store.

6. Define Exploratory Analysis.

- Exploratory Data Analysis (EDA) is an approach to analyzing datasets to summarize their main characteristics, often using visual methods.

- It is done *before* formal modeling to understand the data structure, detect outliers, and test assumptions.

7. Discuss about any two common data visualization techniques.

1. **Scatter Plots:** Used to reveal relationships or correlations between two numerical variables (e.g., Ad Spend vs. Revenue).
2. **Histograms:** Used to show the frequency distribution of a single continuous variable (e.g., age distribution of customers).

8. Retail store removes duplicate customer records collected from multiple branches. Explain what data preprocessing step this is.

- This is **Data Cleaning**.
- Specifically, it falls under *entity resolution* or *deduplication*, where redundant data is identified and removed to ensure that each real-world entity is represented only once.

9. A dataset contains missing values for income. Discuss how data cleaning can handle this problem.

- **Ignore the tuple:** Delete the row (only valid if the dataset is huge).
- **Fill manually:** Time-consuming and not feasible for large data.
- **Use a global constant:** Fill with "Unknown" or "-1".
- **Use central tendency:** Fill with the Mean or Median of the attribute (most common method).

10. An analyst uses a scatter plot to study the relationship between advertising cost and sales. Explain why is this visualization suitable?

- Scatter plots are specifically designed to show the **correlation** between two continuous variables.
- It allows the analyst to visually see if higher advertising costs lead to higher sales (positive correlation), no change, or lower sales, and helps identify outliers.

11. Explain what is the use of a Data Warehouse in organizations?

- It provides a **centralized repository** of historical data from disparate sources.
- It separates analysis workloads from transaction workloads (OLTP), ensuring reporting doesn't slow down daily operations.
- It supports decision-making (DSS) by allowing trend analysis over long periods (e.g., 5-year sales trends).

12. Define OLAP and explain its role in a data warehouse?

- **Definition:** Online Analytical Processing (OLAP) is a technology that allows users to analyze database information from multiple database systems at the same time.
- **Role:** It enables multidimensional analysis (slicing, dicing, drilling down) of the data stored in the warehouse, providing fast answers to complex analytical queries.

13. Differentiate Metadata and Data Mart.

- **Metadata:** "Data about data." It describes the structure, constraints, and meaning of the data (e.g., data types, authorship).
- **Data Mart:** A subset of a data warehouse focused on a specific business line or department (e.g., a Marketing Data Mart).

14. What are facts? Define Fact Table.

- **Facts:** Numeric measurements or metrics that represent business performance (e.g., quantity sold, revenue, profit).
- **Fact Table:** The central table in a star schema that contains these quantitative facts and foreign keys linking to dimension tables.

15. Explain what data warehousing is and its main functions?

- **Definition:** A data warehouse is a subject-oriented, integrated, time-variant, and non-volatile collection of data.
- **Functions:**
  1. **Integration:** Merging data from various sources.
  2. **Storage:** Holding large volumes of historical data.
  3. **Analysis Support:** facilitating complex queries and reporting.

16. Discuss the ETL process in data warehousing?

- **Extract:** Reading data from source systems (databases, logs, XLS).
- **Transform:** Cleaning, formatting, and converting data into a standard schema (e.g., currency conversion, filtering nulls).
- **Load:** Writing the processed data into the target Data Warehouse.

17. Explain the Star Schema in data warehousing and describe its structure with an example.

- **Structure:** The simplest schema. A central **Fact Table** is connected directly to multiple **Dimension Tables**.
- **Example:** A `Sales_Fact` table (containing Revenue, Qty) connected to `Time_Dim`, `Product_Dim`, and `Store_Dim` tables. It looks like a star.

18. What is Snowflake Schema in data warehousing and describe its structure with example.

- **Structure:** An extension of the Star Schema where dimension tables are **normalized** (split into sub-tables).
- **Example:** In a `Product_Dim` table, instead of storing "Category" directly, it links to a separate `Category_Dim` table. This reduces redundancy but increases join complexity.

19. List the difference between OLAP and OLTP.

- **Focus:** OLTP is for day-to-day operations (transactions); OLAP is for decision support (analysis).
- **Data:** OLTP handles current, up-to-date data; OLAP handles historical, summarized data.

- **Users:** OLTP is used by clerks/staff; OLAP is used by managers/analysts.

20. Compare and contrast an operational database with a data warehouse.

- **Operational DB (OLTP):** Optimized for fast read/write of individual records. Normalized to avoid redundancy.
- **Data Warehouse (OLAP):** Optimized for reading large aggregations. Often denormalized (Star Schema) to speed up complex queries.

21. Why is a data mart considered cost-effective compared to a data warehouse?

- **Scope:** It focuses on a single department rather than the entire enterprise.
- **Hardware/Time:** Requires less storage and processing power to build.
- **Implementation:** Faster to deploy, allowing organizations to see ROI quickly before investing in a full enterprise warehouse.

22. List out the various OLAP operations.

1. **Roll-up:** Summarizing data (moving up the hierarchy, e.g., Daily $\rightarrow$ Monthly).
2. **Drill-down:** De-summarizing data (moving down, e.g., Country $\rightarrow$ City).
3. **Slice:** Selecting one dimension (e.g., Sales for "2023").
4. **Dice:** Selecting a sub-cube (e.g., Sales for "2023" AND "Electronics").
5. **Pivot:** Rotating the data axes for a different view.

23. Design a data mart for a specific department (e.g., Finance). Justify selection.

- **Subject:** Finance Department.
- **Data Sources:** ERP System (General Ledger), Payroll System, Invoicing System.
- **Schema Justification:** Use **Star Schema** for simplicity and faster query performance.
- **Fact Table:** `Financial_Transactions` (Amount, Tax).
- **Dimensions:** `Time` (Fiscal Year/Qtr), `Cost_Center` (Dept ID), `Account_Type` (Expense/Revenue).

Section B: Objective / One-Liners (1 Mark Each)

1. **Which of the following is a major application of data mining?**
   - **Answer:** Market Basket Analysis.
2. **Which step comes first in the data mining process?**
   - **Answer:** Data Selection (identifying relevant data sets).
3. **Which step of KDD involves removing noise and inconsistent data?**
   - **Answer:** Data Cleaning.
4. **Data integration refers to?**
   - **Answer:** Combining data from multiple disparate sources into a unified store.
5. **Identify the main goal of data reduction.**

- **Answer:** To reduce the volume of data while preserving its analytical integrity and patterns.

6. **A data analyst plots box plots and scatter plots to detect outliers before model building. Select the step.**
   - **Answer:** Exploratory Data Analysis (EDA).
7. **Before applying any mining algorithm, an organization performs selection, cleaning, transformation, and reduction. Name this stage.**
   - **Answer:** Data Preprocessing.
8. **A telecom company removes noisy call records caused by system error. Which task is this?**
   - **Answer:** Data Cleaning.
9. **Identify what a data warehouse is.**
   - **Answer:** A centralized, subject-oriented repository for storing and analyzing historical data.
10. **Show the primary purpose of a data warehouse.**
    - **Answer:** To facilitate complex data analysis and reporting for decision-making.
11. **Which of the following statements is true about data warehouse?**
    - **Answer:** It consolidates data from various sources specifically for analysis, not transaction processing.
12. **What is ETL in the context of data warehousing?**
    - **Answer:** Extract, Transform, Load.
13. **Which OLAP server type stores summarized data in cubes and detailed data in relational tables?**
    - **Answer:** Hybrid OLAP (HOLAP).
14. **Compare Snowflake schema and Star Schema based on query complexity.**
    - **Answer:** Snowflake Schema has higher query complexity (more joins) than Star Schema.
15. **Which is most suitable for drilled down operation in OLAP?**
    - **Answer:** Viewing detailed monthly sales data after looking at summarized yearly sales.
16. **Which OLAP operation is used to summarize data from lower level to higher level?**
    - **Answer:** Roll-up.
17. **Interpret the purpose of fact table and dimension table in a star schema.**
    - **Answer:** Fact tables store quantitative measures; dimension tables store descriptive attributes/context.
18. **Identify which of the following is a correct ETL activity.**

- **Answer:** Cleaning and standardizing data formats before loading them into the warehouse.

19. **If an OLAP server stores data in multidimensional cubes, what does MOLAP stand for?**
    - **Answer:** Multidimensional Online Analytical Processing.
20. **In a star schema, what does a dimension table contain?**
    - **Answer:** It defines the dimensions, attributes, and hierarchies used to analyze the facts.
21. **A company wants to analyze year-wise sales trends. Which feature best matches the required system?**